

$$3.1) (a) \Rightarrow (b) \text{ má-li } G_1 = G_2 \text{ pak } \langle \vec{u}_i, \vec{u}_j \rangle = \langle \vec{v}_i, \vec{v}_j \rangle \dots \vec{u}_i^T \vec{u}_j = \vec{v}_i^T \vec{v}_j$$

pro $i, j \in \{1, \dots, n\}$

$$Q \vec{u}_i = \vec{v}_i$$

$$Q \vec{u}_j = \vec{v}_j$$

~~neplatí~~

$$\vec{v}_i^T \vec{v}_j = \vec{u}_i^T \underbrace{Q^T Q}_{I_n} \vec{u}_j$$

$$\rightarrow \vec{v}_i^T \vec{v}_j = \vec{u}_i^T \vec{u}_j \quad \checkmark$$

$$(b) \Rightarrow (a)$$

$$G_1 = G_2$$

$$\vec{u}_i^T \vec{u}_j = \vec{v}_i^T \vec{v}_j \dots U^T U = V^T V, \text{ kde } U \text{ je matice } (\vec{u}_1 | \dots | \vec{u}_n)$$

$$V \text{ je matice } (\vec{v}_1 | \dots | \vec{v}_n)$$

chci dokázat $QU = V$, kde Q je ortogonální, tedy: $Q^T Q = I_n$

$$QUU^{-1} = VU^{-1}$$

$$Q = VU^{-1}$$

$$Q^T Q = (VU^{-1})^T VU^{-1}$$

$$Q^T Q = (U^{-1})^T V^T V U^{-1}$$

$$V^T V = U^T U$$

$$Q^T Q = (U^{-1})^T U^T U U^{-1}$$

$$\cancel{Q^T Q} = \underbrace{(U^T)^{-1} U^T}_{I_n} \underbrace{U U^{-1}}_{I_n}$$

$$Q^T Q = I_n \quad \checkmark$$

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